SCHENKTRONICS

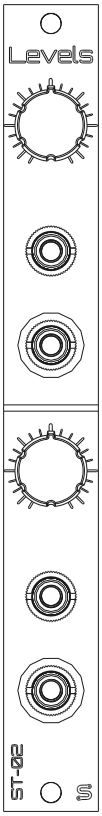
Levels

User Manual

For PCB v1.0 · Updated 2026-10-08

Official version: github.com/Schenktronics/ST02-levels

Overview

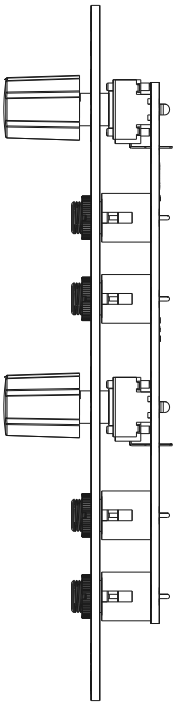


Levels is a 3HP Eurorack utility module with two attenuators. Each one takes a signal at its input and gives you a quieter or smaller copy at its output, set by its knob. In 2 is normalled to In 1, so a single signal can feed both channels at different levels.

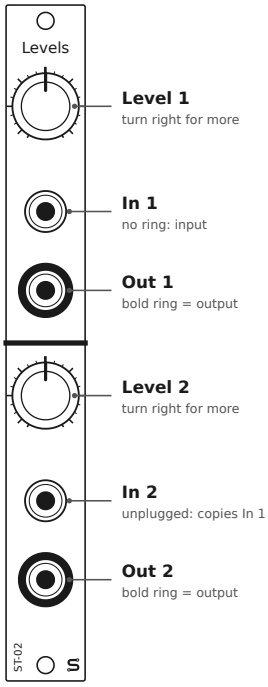
The module is just jacks and potentiometers, so it does not need power.

Specifications

Format	Eurorack, 3HP
Panel	15 × 128.5 mm
Depth	15 mm (skiff-friendly)
Power	None needed
Jacks	4 × 3.5 mm mono (TS): 2 inputs, 2 outputs
Controls	2 × level knobs, 100 kΩ linear potentiometers
Range	Off (fully anticlockwise) to full level (fully clockwise). It can't amplify or invert.
Signals	Audio, CV, gates, and both unipolar and bipolar signals



Panel layout



The panel has two identical channels, separated by the thick line. In each channel, the knob is at the top; below it, the nearer jack is the **input** (unmarked), and the farther jack is the **output**, marked with a **bold ring**.

Using the module

Level knobs

Turn a knob clockwise for more signal. Fully anticlockwise, the output is at 0 V (off). Fully clockwise, the output is close to the full input signal. It is never louder than the input, because the module has no amplifier.

The knob scales the signal toward 0 V. A bipolar LFO swinging ±5 V becomes a smaller swing around 0 V. It isn't shifted or offset.

Normalled input

In 2 is normalled to In 1. When nothing is plugged into In 2, channel 2 gets its signal from In 1. Plug a cable into In 2 to break the normal and use the channels separately.

So with one cable into In 1:

- **Out 1** gives the signal at the Level 1 setting.
- **Out 2** gives the same signal at the Level 2 setting.

This is a quick way to send one modulation source to two places at different depths.

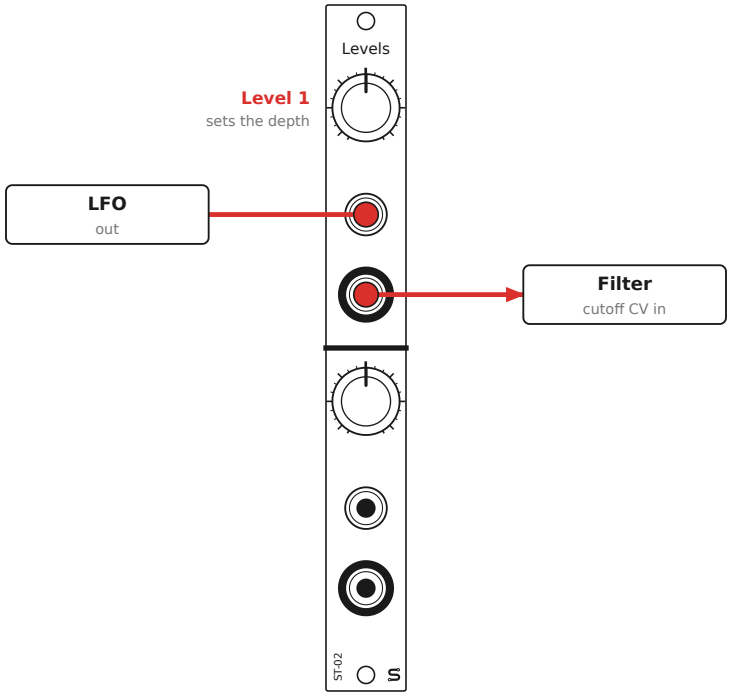
Using it on audio

An attenuator on an audio signal is a volume control. Use it to bring a hot oscillator down before a filter or effect, or to set a manual mix level before a mixer that has no level controls.

Patch examples

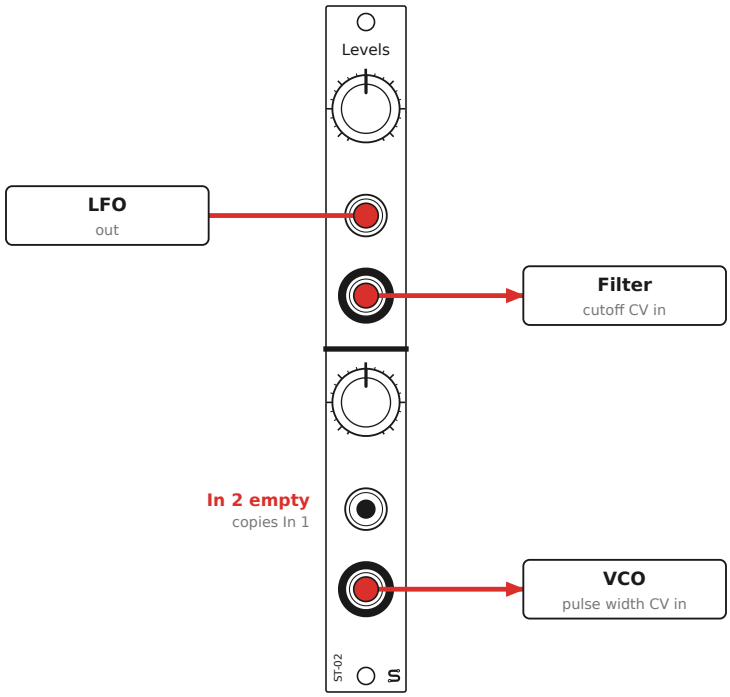
Each drawing shows the module that sends the signal on the left and the modules that receive it on the right.

Set the depth of an LFO



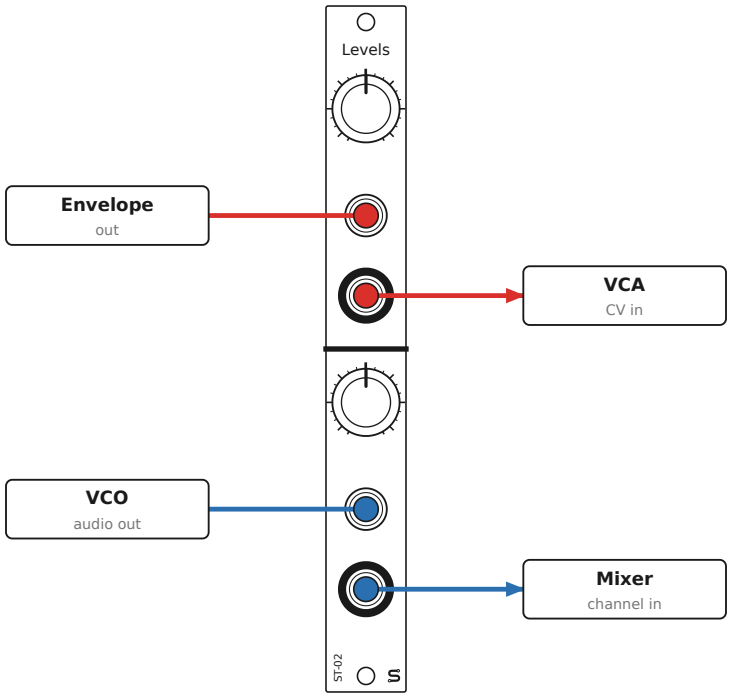
Many filter CV inputs have no attenuator, so a full-size LFO sweeps the filter from closed to wide open. Patch the LFO through channel 1 and use Level 1 to set how far it sweeps.

One LFO at two depths



Leave In 2 empty and the LFO reaches both channels. Here Level 1 sets how much the filter moves and Level 2 sets how much the pulse width moves, from one cable.

Two separate attenuators



With a cable in In 2, the channels are independent. Here channel 1 sets how far an envelope opens a VCA, and channel 2 turns a VCO down before a mixer.

Good practice

- **Patch outputs into the inputs.** Plug sources into the In jacks (unmarked), not the Out jacks (bold ring). A source patched into an Out jack is wired straight to the knob's wiper, so the knob won't work as expected, and turning it fully anticlockwise connects that source directly to ground. Most Eurorack outputs are protected against this, but not all are, so avoid it.
- **Full level is slightly below the input.** A passive attenuator loads the module feeding it and the module it feeds. With typical Eurorack modules, fully clockwise gives a little less than the original signal, and the middle of the knob gives a bit less than half. This is normal for any passive attenuator. For exact levels, use a buffered attenuator.
- **Pitch CV (V/oct):** attenuating a pitch CV changes the intervals, so the oscillator won't track in tune. That can be a creative effect, but it's not a way to transpose.
- **One source into both channels:** when In 2 is empty, In 1 feeds both knobs, so it loads the source a little more. It makes no difference with most modules.
- **Grounds:** the sleeves of all four jacks are wired together.

Installation

1. Power off your case. The module does not need power, but you should never work in a powered case.
2. Place the module in any 3HP space. There is no ribbon cable to connect.
3. Secure it with two M3 rack screws. Do not overtighten them.

Circuit

Each channel is one potentiometer wired as a voltage divider. The input jack's tip goes to one end of the pot, the other end goes to ground, and the wiper goes to the output jack's tip. In 2 (J3) is a switched jack: its normalling contact is wired to the tip of In 1 (J1), so In 1 feeds channel 2 until a plug is inserted in In 2. All sleeves and the pots' mounting lugs are wired to ground. The schematic is in hardware/levels/Levels.sch (https://github.com/Schenktronics/ST02-levels/blob/master/hardware/levels/Levels.sch) (KiCad 5).